

Joint Cooperative Caching and UAV Trajectory Optimization Based on Mobility Prediction in the Internet of Connected Vehicles

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Abstract—In the Internet of Connected Vehicles, caching content frequently requested by users on edge devices can reduce access latency. Particularly in high-traffic density areas, Unmanned Aerial Vehicles (UAVs) can integrate into future cellular networks to enhance the network capacity and meet increased requests. Therefore, we formulate a joint optimization problem of cooperative caching of Base Station (BS) and UAVs and UAV trajectory planning to minimize network latency while considering the limited energy and storage capacity and dynamic vehicles. First, we propose a Temporal-evolving Bipartite Graph Neural Networks (TBGN) model for traveling areas prediction of vehicles. Then, regarding the coupling of optimization variables, we propose an Energy-aware Monte-Carlo Tree Search algorithm to optimize the UAV's service trajectory by predicted spatio-temporal vehicle density. Finally, the optimization problem degenerates into a monotonic submodular function to optimize caching decisions. We utilize real vehicle trajectories for simulations. The results show that the TBGN outperforms other advanced models in terms of mobility prediction accuracy by 7.4%, and the proposed scheme reduces average latency by 16% compared to other schemes.

Index Terms—Internet of Connected Vehicles, unmanned aerial vehicles, edge caching, mobility prediction, trajectory planning.

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I. INTRODUCTION

A. Background and Motivations

WITH the development of emerging intelligent and networked vehicular technology, vehicles can achieve comprehensive network connectivity through advanced communication technologies [1]. However, the growth of these connected vehicles has also led to a dramatic increase in network traffic. Cisco's latest report shows that 5G will generate nearly three times more traffic than the 4G connection [2]. The exponential traffic growth requires expanding the network capacity to meet users' demands. It can be accomplished by deploying dense traffic cells in 5G networks. In addition, edge computing technology is promising to solve the problems caused by the increasing number of devices and large-scale data transmission [3]. However, in dense user areas, the limited cache space of small cells still places a heavy burden on backhaul links. For this reason, cache-enabled unmanned aerial vehicles (UAVs), serving as aerial base stations, are introduced into the network to provide communication services and enhance the network capacity and balance load on the base station (BS) [4].

Moreover, in the Internet of Connected Vehicles (IoCVs), the ground network resources are fixed and have less flexibility [6], [7]. It is challenging to cope with the highly dynamic network environment. UAVs can not only shorten the transmission distance and reduce content access latency but also better adapt to the dynamic environment of IoCVs [5]. The reason is that UAVs have flexibility, deployability, and favorable channel characteristics and can track vehicle movement [8]. Therefore, UAVs can serve as a supplementary component of IoCVs to cooperate with the BS in providing communication and cache resources through flexible dispatch [9].

Motivated by the above requirement, integrating UAVs into future cellular networks to provide communication services has been promising research [10]. Caching frequently requested contents on edge devices effectively addresses network overload. Therefore, choosing cached contents and location is crucial to improve the performance and Quality of Experience (QoE). Although existing works have extensively researched the selection of cached contents and devices, challenges still exist in how UAVs can be integrated into cellular networks to better adapt to the dynamic content demands of IoCVs and reduce latency within the network.

B. Related Works

1) *Popularity-Aware Edge Caching Schemes*: In mobile wireless networks, popularity-aware edge caching schemes are designed to predict the popularity of content requests within the network and select cached contents to reduce content access latency. Thus, the accuracy of content popularity predictions is critical, particularly for heavy traffic loads. Some researchers proposed a learning-based caching scheme. Reference [14] estimated the local content popularity for each BS and predicted the content popularity through a deep regression model in the next period. This scheme only studied the characteristics of the content requests received by the BS to predict the content popularity and did not consider the personalized content needs of the locally served users. Then, to learn the user's personalized request preferences, [15] proposed a Semantic-enhanced Temporal Graph Network to predict content popularity by building a bipartite graph of users and contents. This model enhances popularity prediction performance by incorporating user attention mechanisms and semantic location coding functions. Reference [16] studied the popularity prediction algorithm for content-oriented caching, using the loss of the cache hit ratio to represent cache performance, and proposed a cluster-based popularity prediction framework to determine cached contents. However, the above schemes did not consider how caching decisions should be made in UAV-assisted edge caching. Moreover, [17] investigated an intelligent caching strategy of multiple UAVs deployed for ground users, which studied user request preferences through Latent Dirichlet Allocation (LDA) methods to determine the cached contents of UAVs.

The above scheme only considers time-varying popularity. However, in the high-speed mobile environment of IoCVs, especially for networks with intensive traffic requirements, as vehicles move, the above schemes may not be practical because they require the BS to store all user requests to predict content popularity. In order to more accurately perceive the time-varying requests of served vehicles, we further consider the spatial content popularity distribution based on the location of vehicles' requests.

2) *UAV-Assisted Edge Caching Schemes*: In emergency communication scenarios, UAVs with caching and communication capabilities can support various services. UAV-assisted edge caching can be divided into cache placement and content delivery, where cache placement usually selects the cached contents of UAVs based on the content popularity and the cache capacity of UAVs. To ensure low latency and low power consumption of data delivery, the content delivery process needs to coordinate the location of all ground users and the location of UAVs caching the contents. Cache-enabled UAVs can be used to provide content services to mobile users. Reference [18] proposed user proactive caching technology, which enables the UAV to proactively transmit files to a selected subset of ground nodes and reduce communication energy consumption through ground nodes cooperatively caching all files. During periods of heavy requests, this scheme could significantly increase the load on the network devices. In addition, coordinating the deployment location of UAVs can

optimize service coverage. Reference [19] utilized pedestrian behavior information to predict the distribution of users' content requests and movement patterns, thereby determining the deployment location of UAVs and the cached contents. The scheme did not consider dynamic on-demand scheduling and trajectory design for UAVs when content distribution changes. Moreover, in the strategy of jointly designing UAV trajectory and caching decision, [20] studied the demand heterogeneity and strategy uncertainty and adopted a data-driven approach to ensure the policy confidence level in the case of unknown content distribution information. In fact, the movement of the UAV requires a large amount of energy, and the scheme did not consider optimizing energy efficiency.

Although the above schemes have some in-depth research on UAV caching and content delivery, they are all focused on networks with low user mobility or are temporarily static, which may not be directly applied to IoCVs. Vehicles have higher mobility than ordinary mobile users. The distribution of content requests in local areas is highly correlated with vehicle movement patterns. Therefore, we designed UAVs to provide dynamic caching services that change with vehicle density and request distribution while optimizing energy efficiency.

In the IoCV, UAVs can support UAV-to-vehicle (U2V) communication with high line-of-sight (LoS) probabilities and transmit data with low link interference and pathloss. Reference [6] proposed a content distribution mechanism between UAVs and IoCVs and determined the distribution strategy by designing an alliance game between UAVs and IoCVs. This strategy may cause traffic overload of vehicles and load imbalance challenges in the network. Thus, the dynamic deployment of UAVs needs to be considered. In [12], a proactive file caching and sharing strategy for UAVs was proposed, which combines UAV flight trajectory with data distribution scheduling to improve data distribution efficiency. Reference [11] proposed a content delivery scheme from UAVs to vehicles, studied the service coverage problem of UAVs, and optimized the number of deployed UAVs to serve all vehicles. However, [11] and [12] did not comprehensively consider that the limited energy of UAVs and the optimization of energy consumption for UAV dispatching and service. In order to reduce the energy cost of resource allocation, [13] studied the energy cost of allocating computing and caching resources for mobile users in a central control system and proposed a joint computing and caching optimization framework using an integrated deep deterministic policy gradient algorithm. The framework primarily focused on optimizing the energy cost of computation and caching, neglecting the consideration of UAV services during emergency communications. In addition, for UAV scheduling and caching optimization, [22] studied the driving safety-related content caching strategy of UAV-assisted vehicles, which utilized airship scheduling UAVs and selected cached contents based on users' history request contents and deep Q-network algorithm. However, the deployment and maintenance costs of airships are high, subject to environmental conditions, and require stability and more energy to maintain, and there are still challenges in actual use.

Although the above works research scenarios involving UAV-assisted vehicular networks, research on integrating UAVs into cellular networks for service remains insufficient. Therefore, we explored integrating UAVs into cellular networks for service and enabling them to act as relay nodes for backhaul communication. Moreover, the spatiotemporal variation of content request distribution due to high vehicle mobility and the effect of dynamic traffic load on UAV caching performance should also be considered. In addition, we comprehensively consider that the UAV's on-demand scheduling and limited energy are integrated into the UAV dynamic scheduling process, and the UAV needs to reach any charging station before the energy is exhausted.

C. Contributions

Considering the contradiction between the massive requests and the limited backhaul bandwidth in edge caching in the IoCVs, a model for cooperative caching between UAVs and the BS is proposed. This paper formulates the joint optimization problem of UAV trajectories and UAV-assisted caching with BS. The goal is to minimize network latency while considering the challenges of high vehicle mobility, limited UAV energy, and limited cache capacity. Specifically, we first propose a Temporal-evolving Bipartite Graph Networks model (TBGN) for vehicle mobility prediction. The advantages of this model include: 1) The target area is divided into small areas. The vehicle trajectories are innovatively constructed as bipartite graphs that capture the spatial-temporal relationships of vehicles visiting different areas. Based on this, this model can extract high-order graph node embeddings, leveraging explicit information from inter-domain nodes and implicit relational information from intra-domain nodes; 2) This model does not require accurate real-time vehicle trajectories, which are difficult to obtain and raise potential privacy concerns. It provides a novel approach applicable to scenarios such as task offloading and edge caching; 3) The model can predict vehicles' real-time mobility through offline training and online decision-making. Furthermore, according to the spatio-temporal density prediction of vehicles, we propose an Energy-aware Monte-Carlo Tree Search (EMCTS) algorithm for optimizing UAV trajectories. In addition, by estimating the vehicle's content preferences and request distribution, we further dynamically optimize the cached contents of the BS and UAVs to provide personalized content services. The main contributions of this paper are as follows:

- This paper proposes cooperative caching schemes between a BS and UAVs. The aim is to minimize network latency by formulating the joint optimization problem of cooperative caching and UAV trajectory planning. This research facilitates task coordination and edge caching research in terrestrial vehicular networks.
- We are the first to model vehicle mobility as spatio-temporal relation bipartite graphs and propose a model to predict vehicle mobility, which employs different strategies to capture explicit and implicit information between vehicles and areas. The model utilizes Gate Recurrent Unit (GRU) connection parameters to evolve

the dynamics of network inputs, which can handle changing vehicle sets.

- We propose an EMCTS algorithm for UAV trajectory planning to maximize the service capacity of UAVs. Moreover, we estimate the dynamic content distribution by utilizing mobility prediction and the preference of vehicles to update the cached contents accordingly.
- We use real vehicle trajectory datasets for simulations, and the results show that the proposed TBGN outperforms other advanced mobility prediction models by 7.1% on average. The proposed cooperative caching scheme further reduces the average latency by 16%. This demonstrates that the proposed scheme can effectively improve the network performance.

The rest of this paper is organized as follows. Section II introduces the system model and problem formulation. Section III introduces the proposed cooperative caching scheme. In Section IV, simulation results are presented to evaluate the performance of the proposed scheme. Finally, we conclude our work in Section V.

II. NETWORK MODEL AND PROBLEM FORMULATION

A. System Model

This paper studies UAV-assisted edge caching technology in IoCVs. In the network, an increase in user density often leads to an increase in traffic load within the MBS's service area, which may affect the communication quality of long-distance users. For example, there is a significant event or activity in the city, such as a sports event or New Year's Eve event, which attracts a large number of people and vehicles. Vehicles and mobile users may need access to activity information, traffic situations, software updates, media services, and more, dramatically increasing infrastructure load. The service capabilities of MBS in areas with high business intensity are limited. One reason is that the communication capabilities limit the number of vehicles that MBS can serve. Another reason is that the storage capacity limits the amount of cached contents. This situation can increase the communication capacity and service coverage of the network to improve the communication quality of users. UAVs can play an essential role because of their flexibility. Dynamic deployment and collaboration of UAVs can be used in most emergency communication scenarios to improve users' QoE. By using UAV-assisted edge caching technology, UAVs can be flexibly deployed in areas with intensive user requests or poor communication quality to respond to surges in user requests, thereby reducing content access latency and balancing network load. Therefore, UAVs and MBS collaborative caching can provide content services to more CVs, ensuring a more efficient and responsive service. In the IoCVs, the CV acts as a content requester in the edge caching architecture and can send content requests through the equipped wireless transmission device. The content provider can be an MBS or a UAV with caching and communication capabilities. As content providers, they can proactively cache files that CV may request. Users closer to MBS or UAVs will enjoy better communication quality, and

CV can obtain contents from nearby wireless access points, thereby obtaining higher-quality content services.

The scenario of edge caching for UAV collaborative cellular networks is shown in Fig. 1, where the network consists of an MBS denoted by B , $U = |U|$ UAVs, $V = |V|$ vehicles. Within the communication area of an MBS, there are V vehicles requesting content services. All vehicles are defined as a set $\mathcal{V} = \{1, 2, \dots, V\}$, $v \in \mathcal{V}$. All UAVs are defined as a set $\mathcal{U} = \{1, 2, \dots, U\}$, $u \in \mathcal{U}$. The UAVs fly at altitude H . The MBS is connected to the core network through optical fibers. $\mathcal{F} = \{1, 2, \dots, F\}$ is the set of F files requested by vehicles in the scenario, where $F = |\mathcal{F}|$, $f \in \mathcal{F}$. Without loss of generality, the scenario considers cooperative edge caching services within the area covered by an MBS in the IoCVs. In this model, rotary-wing UAVs equipped with cache units are dispatched into the system to act as mobile BSs, serving ground vehicles together with the MBS. The content units are of the same size, denoted by S_f , the cache capacity of an MBS is S_B , and the cache capacity of UAVs is also the same, denoted by S_U .

We partition the MBS service region into A grids of equal size to facilitate vehicle mobility tracking and UAV dispatching. Each grid is regarded as a small area, and the communication range of a UAV can cover a small area. The center points of areas form a topology map $G' = (\mathcal{A}, \mathcal{E}')$, where $\mathcal{A}$ is the center point set of areas, and the set $\mathcal{E}'$ consists of edges formed by connecting the center points between adjacent areas. The UAV hovers exclusively at the center of an area and flies along the edges connecting the central points of those areas. We divide the endurance time T of a UAV into multiple time slots. The position of UAVs may change in each time slot because they may fly along the edges in successive time slots. The flight trajectory of each UAV is different, and the starting point of the UAV's flight is also different. There are $\hat{K}$ UAV charging stations in the scenario. Note that the UAV needs to arrive at any charging station before it runs out of energy.

B. Communication Model

1) U2V Communication: The architecture will hinder the communication between UAVs and ground vehicles within the urban environment. There are usually two components: LoS and non-line-of-sight (NLoS). According to [24], this paper models the pathloss of UAVs to vehicles by a probabilistic transmission model. The large-scale coefficients of the channel between UAV u and vehicle v in the LoS environment and NLoS environment are expressed as

$$\begin{aligned} \gamma_{\text{LoS}} &= \delta_1 l_{u,v}^{-\tau}, \\ \gamma_{\text{NLoS}} &= \delta_2 \delta_1 l_{u,v}^{-\tau}, \end{aligned} \quad \forall u \in \mathcal{U}, v \in \mathcal{V}, \quad (1)$$

where δ_1 is the pathloss of UAVs related to distance in the LoS environment, $\delta_2 \in (0, 1)$ is the attenuation loss in the NLoS environment, $l_{u,v}$ is the distance between the UAV and the vehicle, and τ is the modeling parameter about the pathloss. The distance $l_{u,v}$ between UAV u and vehicle v is related to the UAV's altitude L and the elevation angle of the UAV and

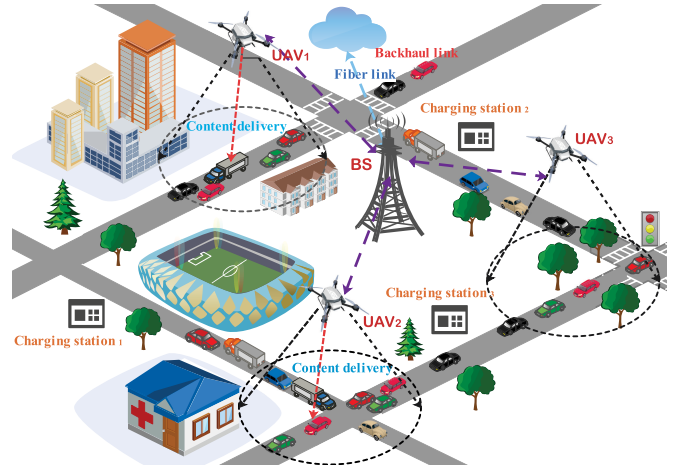


Fig. 1. UAV-assisted Edge Caching in the Internet of Connected Vehicles.

vehicle. It can be expressed as

$$l_{u,v} = \frac{L}{\sin \theta_{u,v}}, \quad \forall u, v. \quad (2)$$

The probability of a LoS link depends on the environment and the elevation angle between UAV u and vehicle v . The probability of a LoS link is expressed as

$$p_{\text{LoS}}(\theta_{u,v}) = \left(1 + \alpha e^{-\beta(\theta_{u,v} - \alpha)}\right)^{-1}, \quad (3)$$

where α and β are environmental factors. Therefore, the probability of an NLoS environment can be expressed as $p_{\text{NLoS}}(\theta_{u,v}) = 1 - p_{\text{LoS}}(\theta_{u,v})$. Then, the channel gain of UAV u and vehicle v is modeled as

$$h_{u,v} = p_{\text{LoS}}(\theta_{u,v}) \gamma_{\text{LoS}} + p_{\text{NLoS}}(\theta_{u,v}) \gamma_{\text{NLoS}}. \quad (4)$$

To simplify the problem, we assume that $\hat{P}$ is the minimum signal power a vehicle receives from a UAV. Only when $h_{u,v} P_U \geq \hat{P}$, UAV u can communicate with vehicle v [25]. P_U is the transmission power of UAVs. Let $\hat{h} = \hat{P}/P_U$, and then the channel gain needs greater than $\hat{h}$. At the same time, the relationship between the UAV's flight altitude and communication range is also obtained. Thus, we have

$$\frac{\delta_1 (H^2 + r_U^2)^{-\frac{\tau}{2}} \left\{ 1 + \alpha \delta_2 \exp \left[-\beta \left(\arctan \frac{H}{r_U} - \alpha \right) \right] \right\}}{\left\{ 1 + \alpha \exp \left[-\beta \left(\arctan \frac{H}{r_U} - \alpha \right) \right] \right\}} \geq \hat{h}. \quad (5)$$

To better illustrate the time-varying locations of UAVs, we partition the target area into small square areas with length r . After determining the UAV's flying altitude, we divide the area according to the UAV's communication radius r_U . Thus, the side is $r = \sqrt{2} r_U$. When a UAV hovers over an area, its communication area is approximately the range of that area. The UAV can cache contents that the vehicle may request in the future, enabling it to transmit the cached contents directly to the mobile vehicle. This approach effectively minimizes transmission latency. According to the achieved signal-to-noise ratio (SNR) levels, the wireless communication area of each UAV is evenly divided into L zones, ordered by their radius corresponding to the distances from the area center. Each zone

forms an annulus area with width l_j from the center. The data transmission rates are calculated based on [26]. A vehicle within an area can achieve a mean transmission rate:

$$\bar{I}_{u,v} = \rho \left(\frac{\sum_{j=1}^L \left(l_j \left[(l_j + l_{j-1})^2 - l_{j-1}^2 \right] \right)}{\left(\sum_{j=1}^L l_j \right)^2} \right), j \in \mathcal{L}, \quad (6)$$

where $l_0 = 0$, and ρ is throughput efficiency factor. Let vehicles share the channel in an area. The maximum number of vehicles that a UAV can simultaneously serve should not exceed

$$N_{u,\max} = \left\lfloor \frac{\bar{I}_{u,v}}{\hat{I}} \right\rfloor, \quad (7)$$

where $\hat{I}$ is a vehicle's requirement for the content transmission rate.

2) The transmission rate of the UAV's backhaul link is

$$I'_u = \min \left\{ W_{B,u} \log \left(1 + \frac{P_B h_{B,u}}{\sigma^2} \right), W_{u,v} \log \left(1 + \frac{P_U h_{u,v}}{\sigma^2} \right) \right\}, \quad (8)$$

where P_B is the transmission power of the MBS, $W_{u,v}$ is the downlink bandwidth allocated by the UAV to the vehicle, $W_{B,u}$ is the bandwidth allocated by the MBS to the UAV, and σ^2 is the noise power.

3) MBS to vehicle (B2V) Communication: Because the vehicle is moving, the channel power gain from the MBS to the vehicle is changing. Allocating real-time power to each vehicle based on channel conditions during different time slots is complex. According to [27], the average pathloss of B2V communication in an area is employed for the power allocation of vehicles. According to the 3rd Generation Partnership Project (3GPP) standard, the pathloss for communication between an MBS and a vehicle located within a specific area is denoted as

$$h_{B,a} = 15.3 + 37.6 \lg(l_{B,a} [\text{m}]). \quad (9)$$

In the absence of UAV-assisted edge caching, the communication rate from MBS B to vehicle v is

$$I_{B,v,t} = \frac{W_B}{\sum_{a \in \mathcal{A}} v_{a,t}} \log \left(1 + \frac{P_{B,a,t} h_{B,a}}{\sigma^2} \right), \quad (10)$$

where $P_{B,a,t}$ is the transmission power from the MBS to vehicles located within area a , W_B is the total bandwidth of the MBS, and σ^2 is the noise power. Let the cellular bandwidth of the MBS be equally shared by the vehicles within its communication range, and the signal power received by the vehicles in each area be equal.

C. File Request Model

Since the vehicles typically drive on urban streets, the vehicle density is usually higher in areas covering urban roads than in other areas, and the probability of content requests is also higher than in other areas. The variation in vehicle density within each area also impacts the probability distribution of content requests for that area. According to [28], the research indicated that the popularity of requested content follows a Zipf distribution derived from the users' records of file requests on the network. It can be considered that 80% of users

request 20% of popular contents, but not all of the popular contents can be cached due to the substantial volume of contents. Users usually have different preferences for different contents. For example, some users pay attention to sports topics, while others like news and entertainment. The content request preferences of vehicles in a small area cannot be obtained only through the statistical distribution of numerous vehicle behaviors. This is attributed to the request preferences of vehicles in a small area changing as the vehicles move in the IoCVs. Therefore, the system needs to predict the expectation of content popularity in the area based on the preferences of vehicles that will enter the area rather than the expected popularity of vehicles in the current area. This enables caching the most valuable contents in a limited space to improve cache utilization.

Given the Zipf-based content popularity, the request distributions still vary spatially and temporally due to the diverse content preferences and high mobility of vehicles. We use statistics to model user preferences based on historical requests. We model the file preference in each area in time slot t . According to the historical file request records, the file request probability matrix is calculated as

$$\mathbf{Q} = \begin{bmatrix} q_{11} & \cdots & q_{1F} \\ \vdots & \ddots & \vdots \\ q_{V1} & \cdots & q_{VF} \end{bmatrix}. \quad (11)$$

Given that each file has a feature vector φ_f , the popularity of file f follows the Zipf distribution. Vehicle v 's request preference for file f is denoted as

$$\psi_v = \sum_{f \in \mathcal{F}} q_{vf} \varphi_f. \quad (12)$$

In area a , the probability of file f being requested in time slot t is

$$q_{a,f,t} = q_f \frac{\sum_{v \in \mathcal{V}} g \left(\frac{\lambda_{v,a,t} \psi_v}{\|\lambda_{v,a,t} \psi_v\|}, \varphi_f \right)}{\sum_{a' \in \mathcal{A}} \sum_{v \in \mathcal{V}} g \left(\frac{\lambda_{v,a',t} \psi_v}{\|\lambda_{v,a',t} \psi_v\|}, \varphi_f \right)}, \quad (13)$$

where $g(\cdot)$ is a kernel function that represents the matching between the vehicle in area a and file f , and $\lambda_{v,a,t}$ is the probability that vehicle v is in area a at t .

D. Energy Consumption Model

The energy consumption of UAVs within the network primarily includes energy consumption during UAV movement, communication-related energy consumption, and hovering energy consumption when UAVs hover to provide content services to vehicles. When the UAV moves with speed μ , according to [29], the energy consumption of a rotary-wing UAV is modeled based on its blade profile power, induced power, and parasite power. The moving energy consumption of the UAV flight per second is defined as

$$E_{mv}(\mu_t) = \tilde{P}_0 \left(1 + \frac{3\mu_t^2}{\mu_0^2} \right) + \tilde{P}_1 \left(\sqrt{1 + \frac{\hat{\mu}_1^4}{4}} - \frac{\hat{\mu}_1^2}{2} \right)^{\frac{1}{2}} + \ell, \quad (14)$$

where $\tilde{P}_0$, $\tilde{P}_1$, and $\tilde{P}_2$ are coefficients representing blade profile power, induced power, and parasite power, respectively; $\hat{\mu}_1 = \frac{\mu_t}{\mu_1}$, μ_0 and μ_1 represent the tip speed of rotor blades and the average rotor-induced speed during hover, respectively; $\ell = \frac{1}{2}\tilde{P}_2\mu_t^3$. These coefficients depend on factors such as the aircraft's weight and air density. It is worth noting that when a UAV hovers to serve a vehicle, the UAV's moving speed is 0. Therefore, the energy consumption of UAV hovering is $E_{hv} = \tilde{P}_0 + \tilde{P}_1$.

When the cache state $\mathbf{c}_{u,t}$ of UAV u is given and the UAV hovers over area $o_{u,t}$ to provide content service for vehicles in time slot t , the communication energy of UAVs is represented as

$$E_{cm}(\mathbf{c}_{u,t}, o_{u,t}) = P_c \cdot \min\{D_{u,t}, \Delta t\}, \quad (15)$$

where P_u is the communication power of UAVs, Δt is the time interval, and $D_{u,t}$ is calculated by Eq. (19).

E. Access Latency

The number of vehicles that need to be served by the MBS in time slot t is derived as

$$n_{B,t} = \sum_{a \in \mathcal{A}, a \notin \mathcal{A}'} v_{a,t}, \quad (16)$$

where $v_{o_{u,t},t}$ is the number of vehicles served by UAV hovering over $o_{u,t}$ in time slot t , $\mathcal{A}'$ is the areas served by UAVs, and $v_{a,t}$ is the number of vehicles of area a in time slot t .

The bandwidth allocated to each vehicle by the cellular BS in time slot t is expressed as

$$W_{B,t} = \frac{W_B}{n_{B,t}}. \quad (17)$$

The latency for vehicles served by an MBS to access contents is denoted as

$$D_{B,t} = \sum_{a \in \mathcal{A}, a \notin \mathcal{A}'} \frac{S_f v_{a,t}}{W_{B,t} \log\left(1 + \frac{P_{B,a,t} h_{B,a}}{\sigma^2}\right)} \sum_{f \in \mathbf{c}_{B,t}} q_{a,f,t}. \quad (18)$$

The average latency for vehicles served by UAV u to access contents is expressed as

$$D_{u,t} = \frac{S_f v_{o_{u,t},t}}{I_u} \sum_{f \in \mathbf{c}_{u,t}} q_{o_{u,t},f,t} + \frac{S_f v_{o_{u,t},t}}{I_u} \left(1 - \sum_{f \in \mathbf{c}_{u,t}} q_{o_{u,t},f,t}\right). \quad (19)$$

The overall network latency is

$$D(\mathbf{C}_u, \mathbf{C}_B, \mathbf{O}) = \sum_{t=1}^T \left(D_{B,t} + \sum_{u=1}^U D_{u,t} \right). \quad (20)$$

F. Problem Formulation

The MBS and UAVs will cache part of popular contents near the vehicles in advance. However, the resources in the network are limited. Under constraints of UAV energy and cache space, a joint optimization problem of cooperative edge caching of UAVs and the MBS is established to minimize network latency. According to the communication model, file

request model, and energy consumption model in Sections II-B, II-C, and II-D, content access latency analysis of the overall network is given in Section II-E. Therefore, the BS cache indicator $\mathbf{C}_B$, the UAV cache indicator $\mathbf{C}_u$ and UAV trajectory $\mathbf{O}$ need to be jointly optimized. The optimization problem $\mathbf{P}$ can be formulated as follows:

$$\min_{\mathbf{C}_u, \mathbf{C}_B, \mathbf{O}} D(\mathbf{C}_u, \mathbf{C}_B, \mathbf{O}) \quad (21)$$

$$\text{s.t.} \quad \sum_{t=1}^T E_{mv}(\mu_t) + E_{cm}(\mathbf{c}_{u,t}, o_{u,t}) \leq E_{\max}, \quad (21a)$$

$$\sum_{f \in \mathcal{F}} c_{u,f} \leq S_U, \quad \forall u \in \mathcal{U}, \quad (21b)$$

$$\sum_{f \in \mathcal{F}} c_{B,f} \leq S_B, \quad (21c)$$

$$o_{u,1}, o_{u,T} \in \{a_0, \dots, a_K\}, (o_{u,t-1}, o_{u,t}) \in \mathcal{E}', \quad (21d)$$

$$c_{u,f} = \{0, 1\}, c_{B,f} = \{0, 1\}, \quad (21e)$$

where $E_{\max}$ is the maximum on-board energy of UAVs, X_u is the maximum cache capacity of UAV u , and X_B is the maximum cache capacity of the MBS. The equation (21) is the optimization objective to minimize content access latency in the network. Constraint (21a) restricts the maximum energy of UAVs. Constraint (21b) limits the cache capacity of UAVs. Constraint (21c) limits the cache capacity of the MBS. Constraint (21d) restrains the trajectory of UAVs, which ensures that the UAVs arrive at any charging station when their energy is exhausted. The UAV only flies along the edge of the area topology graph. Constraint (21e) represents the cache indicator of the UAVs and MBS.

III. A DEEP LEARNING-BASED COLLABORATIVE CACHING DECISION-MAKING SCHEME

In this work, we propose a deep learning-based caching decision-making scheme to solve the collaborative caching problem. We jointly consider the high mobility of vehicles, the dynamics of UAV services, and the constraints of cache space and energy, which will impact the cooperative caching decision. Since the optimization problem $\mathbf{P}$ is non-convex and hard to solve directly, the proposed scheme employs decomposition strategies to transform the original optimization problem into UAV trajectory optimization subproblem $\mathbf{P1}$ and UAVs and MBS cooperative caching subproblem $\mathbf{P2}$. In the scenario, the content request distribution of vehicles changes spatially and temporally with the movement of vehicles. The proposed scheme needs to consider the location where the vehicle sends the content request. The location determines whether the vehicle is provided with caching services by the UAV. Moreover, the distribution of local content requests for vehicles served by the UAV determines the cached contents of the UAV. The spatiotemporal changes in the content request distribution of vehicles in a local area are related to vehicle mobility. The future movement positions of vehicles will influence the trajectory optimization of UAVs and the distribution of content requests in different areas, thereby affecting caching decisions. Therefore, the proposed scheme considers vehicle mobility to enhance optimization performance and simplify the problem.

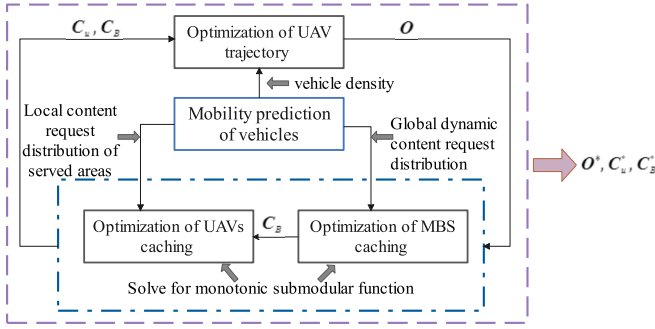


Fig. 2. Overall optimization framework.

Based on the above analysis, the proposed scheme optimizes UAV trajectory and caching decisions by predicting vehicle mobility. The overall optimization framework is shown in Fig. 2. Specifically, we first divide the area according to the communication range of a UAV. Then, we propose a TBGN model to predict vehicle mobility based on the time-varying visiting relationships. It optimizes the service area of UAVs by predicting the future movement patterns of vehicles, which will be elaborated in Section III-A and Section III-B. Based on the predicted spatiotemporal density of vehicles, the UAV selects a path that satisfies more vehicle demands under service capacity and energy constraints. After training the vehicle mobility prediction model, we estimate the content request distribution in the served areas by the UAV based on the vehicle's mobility prediction and request preference to optimize the cached contents. According to the content request distribution estimation and UAV service trajectory, the UAVs and MBS cache optimization problems degenerate into a monotonic submodular function solution. Concretely, in order to better optimize the problem **P**, we decompose the original optimization problem into two optimization problems: subproblem **P1** and subproblem **P2**. The **P1** that focuses on optimizing the UAV trajectory **O**:

$$\begin{aligned} \min_{\mathbf{O}} \quad & D(\mathbf{C}_u, \mathbf{C}_B, \mathbf{O}) \\ \text{s.t.} \quad & (21a), (21d), (21e). \end{aligned} \quad (22)$$

The cooperative edge caching subproblem denoted as **P2** that focuses on optimizing the cache indicator $\mathbf{C}_u$ of UAVs and cache indicator $\mathbf{C}_B$ of the MBS:

$$\begin{aligned} \min_{\mathbf{C}_u, \mathbf{C}_B} \quad & D(\mathbf{C}_u, \mathbf{C}_B, \mathbf{O}) \\ \text{s.t.} \quad & (21b), (21c). \end{aligned} \quad (23)$$

A. Mobility Prediction

In this section, we propose a TBGN model for vehicle mobility prediction. The scheme predicts the moving area of vehicles instead of the real-time trajectory for the following reasons: 1) The real-time monitoring of the vehicles' location is costly and unrealistic. Moreover, the real-time uploading of precise location information by vehicles will raise privacy concerns; 2) To update the cache, edge nodes are not required for real-time vehicle trajectory tracking. Instead, they only

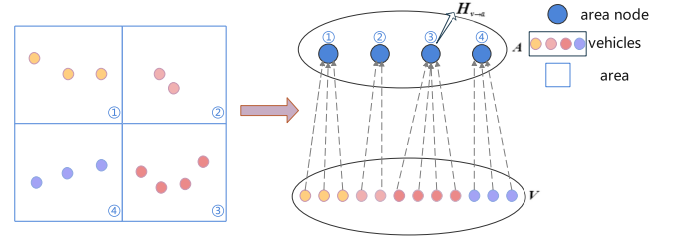


Fig. 3. The relationship between vehicles and areas is transformed into a Bipartite Graph.

need to update the cache by predicting the vehicles' located area within a given time slot; 3) Compared with the research on vehicle mobility of autonomous driving, edge caching reduces the need for accurate vehicle trajectories.

Based on the above analysis, in the designed vehicle mobility prediction model, the prediction of vehicle movements between different areas is accomplished by abstracting the visiting relationship between vehicles and areas as time-varying bipartite graphs. Specifically, the mobility prediction can be transformed into solving the problem of their visiting prediction. The visiting relationship can be converted into a bipartite graph for each time slot, as shown in Fig. 3. If a vehicle is located in area a in time slot t , there is a link exists between the vehicle and the area. Vehicles are separately connected to nodes that represent the corresponding areas where they are located. Therefore, according to the vehicle records in each period, we design a time-varying bipartite graph network $G = (\mathcal{V}, \mathcal{A}, \mathcal{E}, \mathcal{T})$, which is defined based on the vehicles' visiting records of different areas in each time slot. In G , $\mathcal{T}$ is the total time slot, $\mathcal{V}$ is the set of vehicles in each time slot, $\mathcal{A}$ is the set of areas, and $\mathcal{E}$ is the set of visiting relationships in each time slot. Each time slot corresponds to a graph representation that changes with the relationship, which enables us to model and learn the time series information in the graph data. Moreover, as vehicles move, they may leave or disconnect from this network, leading to a dynamically changing set of vehicles within the system.

Our proposed model first extracts node representations based on a cascade bipartite graph neural network to learn the visiting relationship [30]. In this process, the bipartite graph neural network is used to explore the relationship between nodes, and different strategies are adopted to obtain node information, including explicit and implicit information [30]. Different types of nodes can be regarded as different domains. The bipartite graph neural network consists of two modules, including information aggregation of inter-domain nodes and message alignment of intra-domain nodes, which extract explicit and implicit information about nodes, respectively. The model obtains explicit information between different types of nodes based on the information aggregation method, propagates information between two types of nodes on the visiting relationship graph, and encodes the information of inter-domain nodes into node features. The model uses a message alignment mechanism to extract implicit information between nodes of the same type, fusing the information obtained from the graph with the features of the current

domain nodes. At the same time, the adversarial model minimizes the differences between input features and explicit relational representations.

Specifically, we can transform the bipartite graph into a visiting relationship matrix. Let $\mathbf{R}$ denote a visiting relationship matrix between vehicles and areas, the value of element $R_{v,a}^t$ in $\mathbf{R}$ is either 1 or 0. Notably, the visiting relationships only exist between nodes of different types. If $R_{v,a}^t = 1$, it means that vehicle v is located in area a in time slot t ; otherwise, vehicle v is not in area a . During a time slot, multiple vehicles can be present in the same area, and a vehicle can only exist in one area. Node representation is extracted through a bipartite graph neural network, where node representation learning is to learn a low-dimensional representation of each node based on the node attribute information and visiting relationship matrix. During the information extraction process between nodes, the information aggregation method aims to preserve the network structure and learn the attribute proximity of nodes by aggregating neighborhood information. Take set $\mathcal{A}$ to aggregate information from set $\mathcal{V}$ as an example, it can be achieved by aggregating the adjacency matrix of a and the input feature of the neighborhood X_v to obtain information from the neighborhood $H_{v \rightarrow a,k}$. Therefore, the information of other neighborhood nodes can be extracted from the explicit relationship between nodes of different types. In this work, explicit relational representations of nodes in each domain are obtained through two information flows, i.e., the aggregation process from set $\mathcal{V}$ to set $\mathcal{A}$ and from set $\mathcal{A}$ to set $\mathcal{V}$ can be expressed as:

$$\begin{aligned} H_{v \rightarrow a,k} &= \sigma(\hat{R}_a H_{v,k} W_{a,k}), \\ H_{a \rightarrow v,k} &= \sigma(\hat{R}_v H_{a,k} W_{v,k}), \end{aligned} \quad (24)$$

where $H_{v \rightarrow a,k}$ and $H_{a \rightarrow v,k}$ are information representations of k -th layer aggregated from vehicle set $\mathcal{V}$ and area set $\mathcal{A}$, respectively, $v \in \mathcal{V}$, $a \in \mathcal{A}$; $H_{v,k}$ and $H_{a,k}$ are the hidden layer representations of vehicle v and area a at k -th layer, respectively; k is the number of layers; $\hat{R}_a$ and $\hat{R}_v$ are bipartite graph adjacency matrices of the vehicle and area in R , respectively; $W_{v,k}$ and $W_{a,k}$ are the weight of vehicle v and area a at the k -th layer, respectively; $H_{a,0} = X_a$ and $H_{v,0} = X_v$, where X_v and X_a are the original input features of vehicle v and area a , respectively. In order to fuse the explicit information obtained from the information of neighborhood nodes with the input feature of original domain nodes, a discriminator needs to be trained to distinguish them. Specifically, intra-domain node message alignment uses discriminators to improve the ability to recognize representations of neighborhood aggregation features and input features (hidden layer attributes). Information aggregation methods for inter-domain nodes are trained as generators to prevent the discriminator from making accurate predictions. The model implements implicit message alignment of nodes with the same domain through adversarial learning:

$$\begin{aligned} L(v) &= L_{\text{adv}}(H_{a \rightarrow v,k}, H_{v,k}), \\ L(a) &= L_{\text{adv}}(H_{v \rightarrow a,k}, H_{a,k}), \end{aligned} \quad (25)$$

where $L_{\text{adv}}()$ is the adversarial loss function. The implicit information of nodes in the same domain is combined through the self-supervised method, which is trained to obtain low-dimensional representations $H_{a,k}$ and $H_{v,k}$ of different nodes. The model exploits both network topology information and node features from two domains. Taking vehicle nodes as an example, the extracted node representation $H_{v,k}$ not only incorporates the relationship information of the neighborhood but also incorporates the information contained in original input X_v by minimizing the adversarial loss function. Specifically, the adversarial loss function uses two methods to align $H_{a \rightarrow v}$ with H_v . The discriminator is trained to distinguish between feature vectors randomly sampled from $H_{a \rightarrow v}$ and H_v to discriminate different feature representations accurately. Then, the generator aggregates the encoded representation $H_{a \rightarrow v}$ to H_v , preventing accurate discrimination by the discriminator. They are trained to converge to Nash equilibrium and learn representation H_v . Similarly, the training of area nodes follows the same process. The loss function for the discriminator is

$$\begin{aligned} L_{\text{Dis}}(\xi|\eta) &= \frac{1}{V} \sum_{i=1}^V \log P_{\xi,\eta}(\text{sou} = 0 | H_{v_i}) \\ &\quad - \frac{1}{A} \sum_{i=1}^A \log P_{\xi,\eta}(\text{sou} = 1 | H_{a \rightarrow v_i}), \end{aligned} \quad (26)$$

where the discriminator parameters are defined as ξ , the generator parameters are defined as η , $P_{\xi,\eta}(\text{sou} = 0 | x)$ denotes the probability that input feature vector x is from domain H_v , and conversely, $\text{sou} = 1$ means that x is from the target domain $H_{a \rightarrow v}$. The loss function for the generator is

$$L_{\text{Gen}}(\eta|\xi) = \frac{1}{A} \sum_{i=1}^A \log P_{\xi,\eta}(\text{sou} = 0 | H_{a \rightarrow v_i}). \quad (27)$$

The goal is to learn the embedding representation H_v and H_a for nodes in sets $\mathcal{V}$ and $\mathcal{A}$, respectively. Finally, by concatenating the output results of each hidden layer, the final node representation of the time slot is obtained.

We focus on predicting the future visiting relationships between vehicles and areas. The vehicle movement leads to changes in their visiting relationships, generating a series of visiting relationship graphs with a temporal relationship. In future graph representation learning, recurrent neural networks can be introduced to model sequence data and learn the time-varying information of node features extracted from bipartite graphs. However, if learning the temporal dynamics of node information, the methods require that the nodes (vehicles) always exist during the system cycle. Thus, these methods are more challenging to apply for vehicles with high mobility in the IoCVs. This is because the movement of vehicles may cause them to lose connection with the network in that area (network access point switching), so the vehicle set of the system is dynamic. Therefore, to flexibly deal with dynamic nodes in bipartite graphs, the proposed model captures the dynamics of graph representation by learning the changes in network parameters and input features. The model utilizes a Gate Recurrent Unit (GRU) to evolve weight

parameters that vary with input features and time instead of learning the temporal dynamic features of each node to capture dynamic information. In this case, the vehicle nodes in the graphs can freely exist in different system time slots. The model design is shown in Fig. 4. The proposed model regards the same layer parameters of each time slot as a sequence and then utilizes GRU to learn the time dependence between the weights. Then, the learned weight parameters are input into the bipartite graph neural network, which combines the relationship matrix and original features to extract the k -th layer embedding representation of nodes. The proposed model is denoted by

$$\begin{aligned} W_{v,k}^t &= f_{\text{GRU}} \left(H_{v,k-1}^t, W_{v,k}^{t-1} \right), \\ H_{v,k}^t &= f_{\text{BGN}} \left(R_v^t, H_{a,k}^t, X_v^t, W_{v,k}^t \right), \\ W_{a,k}^t &= f_{\text{GRU}} \left(H_{a,k-1}^t, W_{a,k}^{t-1} \right), \\ H_{a,k}^t &= f_{\text{BGN}} \left(R_a^t, H_{v,k}^t, X_a^t, W_{a,k}^t \right), \end{aligned} \quad (28)$$

where $H_{v,k-1}^t$ and $H_{a,k-1}^t$ are the embedding representation of the $(k-1)$ -th layer about vehicles and areas in time slot t , respectively; $W_{v,k-1}^{t-1}$ and $W_{a,k-1}^{t-1}$ are the $(k-1)$ -th layer parameters of vehicles and areas in time slot $t-1$, respectively; $W_{v,k}^t$ and $W_{a,k}^t$ are the k -th layer parameters of vehicles and areas in time slot t , respectively; f_{BGN} is a bipartite graph neural network model; f_{GRU} is the GRU model. Take the example of extracting vehicle representations, f_{GRU} can be expressed as:

$$\begin{aligned} I_{k-1}^t &= \sigma \left(\omega'_I \cdot \left[W_{v,k-1}^{t-1}, H_{v,k-1}^t \right] \right), \\ r_{k-1}^t &= \sigma \left(\omega'_r \cdot \left[W_{v,k-1}^{t-1}, H_{v,k-1}^t \right] \right), \\ \tilde{W}_{v,k-1}^t &= \tanh \left(\omega' \cdot \left[r_{k-1}^t W_{v,k-1}^{t-1}, H_{v,k-1}^t \right] \right), \\ W_{v,k-1}^t &= (1 - I_{k-1}^t) \cdot W_{v,k-1}^{t-1} + I_{k-1}^t \cdot \tilde{W}_{v,k-1}^t, \end{aligned} \quad (29)$$

where $\sigma(\cdot)$ is the sigmoid function; ω'_I , ω'_r and ω' are the weight parameter of the model; I and r are the update gate and reset gate; $\tilde{W}_{v,k-1}^t$ is the updated weight; $\tanh(\cdot)$ is the tanh function; $W_{v,k-1}^{t-1}$ is the weight of the $(k-1)$ -th layer about the vehicle in time slot $t-1$; $H_{v,k-1}^t$ is a embedding node representation of the $(k-1)$ -th layer about the vehicle in time slot t .

B. UAV Trajectory Optimization

Maximizing the cache hit rate for content requested by vehicles from nearby edge nodes can reduce the average latency for vehicles to acquire contents. At the same time, more vehicles served by UAVs can also effectively reduce latency. The trajectory of UAVs needs to be optimized to minimize system access latency so that it can adapt to changes in the spatiotemporal distribution of content requests due to vehicle movement. However, the future distribution of requests is unknown, so making future decisions is challenging. Moreover, UAV energy is limited, and the benefits and costs of UAV movement and scheduling should be considered. Monte Carlo Tree Search (MCTS) is a search algorithm for decision problems that can be used to make decisions without complete

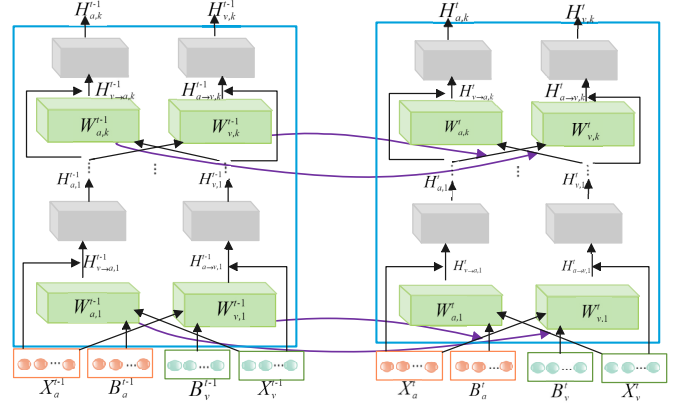


Fig. 4. The Temporal-evolving Bipartite Graph Networks model.

information about the system [31]. It constructs the decision process into a search tree, evaluates different decision paths to the termination state by simulating multiple random events, and selects the optimal decision according to the evaluation results. The MCTS is a $\langle \text{state}, \text{action} \rangle$ driven method to determine the optimal action decision via stochastic simulations. It can make real-time decisions about UAV movement based on feedback on the current vehicle and environment state. The UAV can obtain information in the current area based on the vehicle's movement prediction and select a trajectory that provides more service. The mobility prediction of vehicles in the future time slots can be obtained by Section III-A. The content request distribution prediction can be obtained from Section II-C.

Therefore, we propose an Energy-aware Monte Carlo Tree Search algorithm to address UAV trajectory optimization subproblem **P1**. Based on the MCTS algorithm, we construct an energy-aware search tree by evaluating feasible actions and energy consumption under different observations based on the MCTS algorithm. Each layer of the search tree corresponds to a different time slot. The tree nodes of each layer represent the UAV's state at different time slots. The optimized path selection should ensure that the UAVs return to the nearest charging station before running out of energy and provide more content services. The status information contained in each tree node includes UAV hovering point O_i , UAV action Z_i , reward value M_i , and the number of visits X_i to node i , $i \in \mathcal{N}$. To better calculate the energy consumed by moving UAVs, UAVs can only move from one area to an adjacent area in contiguous time slots. Therefore, in the search tree, the current state (node) of the UAV is transferred to (connected to) the next state (child node) in the next time slot (layer) by selectable actions. For example, if the UAV hovers at point $(x_{u,t-1}, y_{u,t-1})$ in time slot $t-1$, its optional next moving points are: $\{(x_{u,t-1}, y_{u,t-1}), (x_{u,t-1} \pm r, y_{u,t-1}), (x_{u,t-1}, y_{u,t-1} \pm r), (x_{u,t-1} \pm r, y_{u,t-1} \pm r)\}$. The tree's optimal search path is considered as the UAV's action trajectory in each time slot. The initial cached contents of UAVs can first be determined based on the vehicles' initial location and content preferences. The number of nodes in the search tree is $|\mathcal{N}|$. The UAV's attributes comprise the UAV's

service area, the number of vehicles served, and residual energy. The search and iteration of nodes follow the steps:

1) *Selection*: Starting with the UAV's initial state as the root node, the Upper Confidence Bounds (UCB) algorithm recursively searches from the root node to the current child node (not a fully expanded node or leaf node). This recursive process is the UAV beginning from its current initial hovering point and calculating the rewards associated with different actions based on the mobility prediction of vehicles and UAV's attributes. Then, the UAV flies the selected candidate hovering point with the highest UCB value. UCB can well balance the relationship between exploration and exploitation, which can be expressed as

$$\text{UCB}(n_t, z) = \arg \max_z \left(\frac{M(n_t, z)}{X(n_t, z)} + \omega \sqrt{\frac{2 \ln X(n_t)}{X(n_t, z)}} \right), \quad (30)$$

where z is an action of the UAV, the constant ω is the trade-off factor between exploration and exploitation, $M(n_t, z)$ is the sum of reward values of node n_t , $X(n_t, z)$ is the number of visits to the child node, and $X(n_t)$ is the number of visits to the parent node.

2) *Expansion*: In this step, the algorithm expands the current leaf node of the tree. If the current node (the one with the highest UCB value) has both non-terminal state and unexpanded child nodes, the algorithm will select the most promising unexplored node and an unexpanded action to create and initialize a new child node. The UAV explores unobserved action z in the current hovering point to reach hovering point $O(n_{t+1}, z)$ of the next child node.

3) *Simulation*: Monte Carlo simulation evaluation is performed from the current leaf node. The UAV's actions are randomly selected and executed in each time slot until it reaches a low energy state (the residual energy can only support moving to a nearby charging station). At this point, the UAV's service is terminated. During the process, the UAV will move from its current area to a randomly chosen nearby candidate area. Note that the simulation process only generates the corresponding tree node's virtual state and updates the tree node's state statistics through the stochastic optimization process to explore the optimal trajectory of the UAV. The termination condition of the simulation process is

$$\sum_{t=0}^T E_{\text{cm},t} + E_{\text{mv},t} \geq E_{\text{ava},t}, \quad (31)$$

where $E_{\text{cm},t}$ and $E_{\text{mv},t}$ are the communication energy consumption and mobile energy consumption in time slot t , respectively; $E_{\text{ava},t}$ is the available energy, calculated as $E_{\text{max}} - \min \{E_{\text{mv},t}(a_i), i \in \{0, \dots, \hat{K}\}\}$; $E_{\text{mv},t}(a_i)$ is the energy consumed by moving from the current area to charging station a_i .

4) *Backpropagation*: Calculate the reward value generated by the simulation. The reward function obtained according to the UAV's action selection for each time slot is defined as

$$M_t = \frac{E_{\text{cm},t}}{E_{\text{max}}} + \frac{E_{\text{cm},t}}{E_{\text{mv},t}}. \quad (32)$$

C. Caching Optimization

In general, caching the most popular files can effectively reduce backhaul load, attributed to the long-tail distribution of file popularity. However, the global popularity of files is determined by popular trends derived from content requests of numerous users, which might not apply to the localized distribution evaluation of requests in a specific area. Moreover, in our scenario, the UAV serves a small-scale area. Unlike typical users, vehicles have high mobility, resulting in constant changes in vehicle density and locations. Therefore, the cached contents need to change dynamically according to the future content request distribution in the specific area. According to the mobility prediction and preference of vehicles, the distribution of content requests in a specific area can be estimated to match dynamic content popularity within a local area more effectively. We have modeled vehicle preferences and file attributes to estimate the file request probability in Section II-C.

To solve cooperative edge caching subproblem **P2**, the cache optimization goal is to improve the cache hit rate to minimize the content request latency. Therefore, we only focus on predicting vehicle behavior patterns related to content requests. Predicting the time and location for vehicle request files can be categorized into two types of behavior patterns: temporal behavior patterns and spatial behavior patterns. Temporally, we assume the vehicle requests content randomly according to their preferences in each time interval. Spatially, we predict the areas where vehicles initiate file requests via the TBGN model proposed in Section III-A.

In addition, the vehicle's request connection strategy is critical in the cache deployment design. On the one hand, the signal strength of wireless transmission is affected by the distance between devices. On the other hand, when a vehicle initiates a request for contents that a UAV has not cached, there will be no caching benefit. In the connection strategy, when the vehicle is located in the communication range of the UAV, it is served by the nearest UAV. Otherwise, the vehicle is served by the MBS. Therefore, there is no coupling relationship for the content deployment between UAVs. Correspondingly, it is only necessary to independently optimize the cached contents for each UAV based on the content request distribution of its served area. According to the definition of the file request model in Eq. (13), we can obtain the probability $q_{a,f,t}$ that the vehicle requests file f in area a in time slot t . In Section III-B, the trajectory of UAVs has been optimized according to the initial cache. The optimization problem **P2** can degenerate into the problems shown in Eq. (33) and Eq. (34):

$$\begin{aligned} c_u &= \arg \max_{c_u} \sum_{f \in \mathcal{F}} c_{u,f} q_{a,f,t} \\ \text{s.t.} \quad &\sum_{f=1}^F c_{u,f} \leq S_U, \end{aligned} \quad (33)$$

where $c_{u,f}$ is the indicator of file f cache by the UAV, and c_u is the cache indicator vector of the UAV. When a vehicle enters the communication range of the UAV and its requested file is cached by the UAV, the cached contents of the MBS can be optimized based on the file request distribution

of the remaining areas not served by the UAV. The cache optimization problem of the MBS can also be reformulated as an optimization problem of a monotonic submodular function

$$\begin{aligned} c_B &= \arg \max_{c_B} \sum_{f \in \mathcal{F}} \sum_{a' \in \mathcal{A}'} c_{B,f} q_{a',f,t} \\ \text{s.t. } \sum_{f=1}^F c_{B,f} &\leq S_B, \end{aligned} \quad (34)$$

where $\mathcal{A}'$ is the area not served by the UAV, $c_{B,f}$ is the indicator of file f cache by the MBS, and c_B is the cache indicator vector of the MBS. This problem is NP-complete and can usually be solved using a greedy algorithm. The element that contributes the largest marginal increment to the objective function is selected at each iteration.

IV. PERFORMANCE EVALUATION

A. Simulation Parameters

In this section, the real-world Shenzhen vehicle trajectory dataset [32] is used for simulations to verify and analyze the effectiveness of our scheme. The dataset contains information about vehicle trajectories in Shenzhen for one day, including latitude and longitude, passenger status, and speed. We filtered and analyzed the data to evaluate the accuracy of the proposed vehicle mobility prediction model. We preprocessed data by choosing vehicle data from a $2 \text{ km} \times 2 \text{ km}$ area and a specific period with the greatest number of vehicles. We converted the coordinate information (latitude and longitude) acquired by the vehicle at different time intervals into the corresponding area number. Then, we processed the vehicle data by time alignment and shortest path filling methods to obtain the trajectory record every 15 seconds. At the same time, we excluded vehicles with fewer than 50 trajectory records to reduce noise and anomaly data.

In the simulation of vehicle content requests, acquiring accurate user information will be challenging because wireless network operators usually hold it. Therefore, we assume that the popularity of files follows a Zipf distribution with an exponent of 0.7. Regarding UAV energy consumption, some system parameters $\tilde{P}_0$, $\tilde{P}_1$, and $\tilde{P}_2$ are set to 79.85 J/s, 88.63 J/s, 0.018 kg/m, u_0 , u_1 are set to 120 m/s, 4.03 m/s, and the time interval is 15 s. Then, we compare the proposed TBGN with other trajectory prediction models to verify its accuracy (ACC). Moreover, we also simulate and analyze the proposed cooperative caching scheme and UAV trajectory planning algorithm to analyze their impact on content access latency under different system parameters. The parameter settings in the simulations are as shown in Table I. We compared the following mobility prediction models: ContextVAE [34], GATraj [33], STGCNN [35] and Long Short-Term Memory (LSTM).

B. Evaluation of Mobility Prediction

Fig. 5 shows the comparison between the TBGN model and other models in the accuracy of vehicle mobility prediction. As shown in Fig. 5, the prediction accuracy of TBGN is better than other models in most time slots. As can be seen

TABLE I
SIMULATION PARAMETERS

Parameter	Value
The pathloss parameter of UAVs in the LoS environment, δ_1	7×10^{-5}
The attenuation loss parameter in the NLoS environment, δ_2	0.01
The modeling parameter about the pathloss, τ	2
The bandwidth of UAVs, W_u	5 MHz
The bandwidth of the MBS, W_B	50 MHz
UAV transmission power, P_u	33 dBm
MBS transmission power, P_B	43 dBm
Noise power, σ^2	-174 dBm
The size of a file, S_f	200 KB
The cache space of UAVs, S_U	40 MB
The cache space of the MBS, S_B	80 MB
Total energy of UAVs, $E_{\max}$	50 KJ
The speed of UAVs, μ	50 km/h

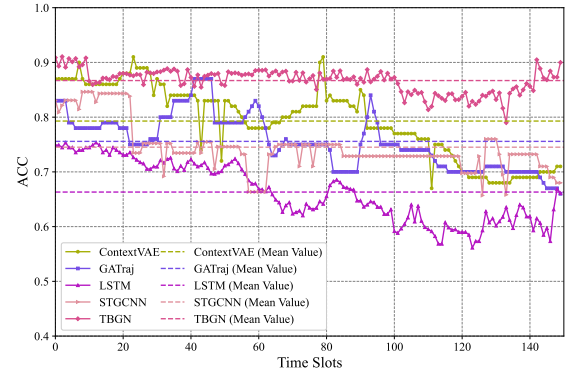


Fig. 5. Vehicle mobility prediction accuracy of different algorithms.

from the mean value lines, compared with the second-ranked ContextVAE model, the prediction accuracy is improved by 7.4% on average. TBGN extracts the dynamic interaction between the agent and the environment. It can be considered that the interaction information between the agent and environment may be more critical than the observation coding that extracts the environment context and the agent state separately in ContextVAE for traveling area prediction. Moreover, it is worth mentioning that there are fluctuations in the performance after the 100-th time slot, which may be affected by the dataset's characteristics. From the figure, it can be observed that our proposed model shows higher stability compared to other models. This may be because of the dynamic nature of the vehicle set within each time slot in the network, allowing for the possibility that not every vehicle exists throughout the training process. At the same time, it indicates that TBGN can effectively manage dynamic network changes by evolving its parameters in response to variations in input features. However, compared with other algorithms, GATraj and STGCNN have similar and slightly lower performance, which indicates that the model only considers the interaction modeling between agents and may not be well adapted to the dynamic environment, and the performance can be improved by using scenario information. In particular, our model outperforms the LSTM by an average of 20.2%. It can be noted that the prediction accuracy of the LSTM gradually decreases over time. This downward trend may be because LSTM may not

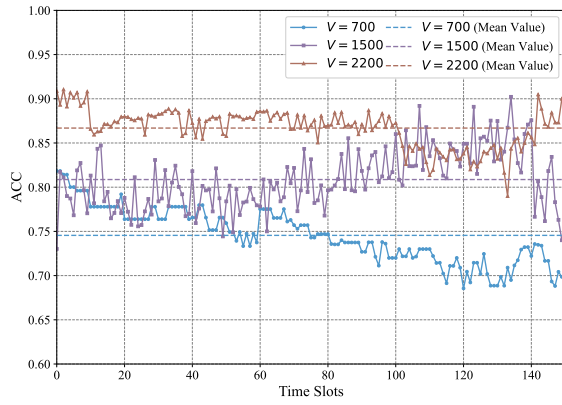


Fig. 6. The influence of the number of vehicles on mobility prediction accuracy.

effectively learn the association relationship between newly arriving vehicles.

Fig. 6 shows the influence of the number of vehicles on mobility prediction accuracy. From the figure, the stability and performance of our model improve as the number of vehicles increases. When the number of vehicles increases from 700 to 2200, an average 12.1% improvement in performance is observed according to the mean value lines. Moreover, the curve of 700 vehicles has the lowest performance, and the fluctuation of the prediction results is more obvious and has a decreasing trend. This may be attributed to the limited number of vehicles and vehicle data with fewer trajectory records in the data, which may not be able to fully capture and learn from additional information, resulting in low robustness. Moreover, it can be observed that when the number of vehicles exceeds 1500, the performance gap diminishes as time increases. This is because we removed some incomplete data with fewer trajectory records from the dataset of 1500 vehicles, leading to reduced interference from the missing data. By observing the distribution of data on the mean value lines, it can be noted that the greater the number of vehicles, the better and more stable the performance. This indicates that increasing the vehicle's data and filtering out incomplete data can improve performance. It is worth noting that there are still some data fluctuations after the 100-th time slot. This may be because of the effect of dataset differences.

Fig. 7 shows the impact of the vehicle mobility prediction models on the content access latency. We can observe that our model exhibits the best performance from the figure. It can be considered that mobility prediction accuracy is crucial for performance improvements, possibly because vehicle mobility predictions affect estimates of content request distribution, thus affecting UAVs' trajectory planning and caching schemes. Moreover, it can be noted that compared with the suboptimal model, our model reduced by 10.3 % in latency. This indicates that the graph-based prediction and vehicle visiting relationship are more effective for vehicle density estimation, which can be used to optimize UAVs' service area and cached contents. Therefore, improving latency performance is closely associated with higher accuracy in mobility prediction. Furthermore, it can be observed that latency tends to increase over time, potentially due to decreased prediction accuracy caused

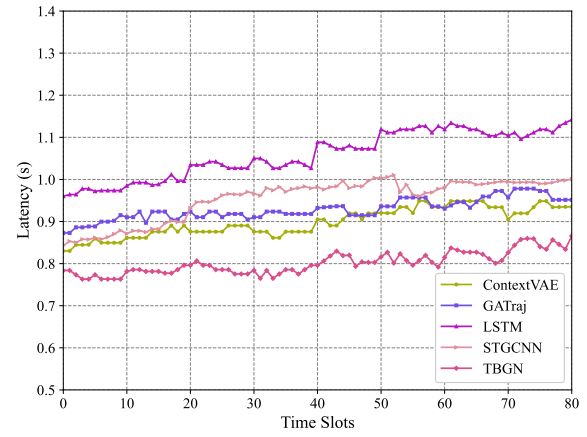


Fig. 7. Content access latency of different algorithms.

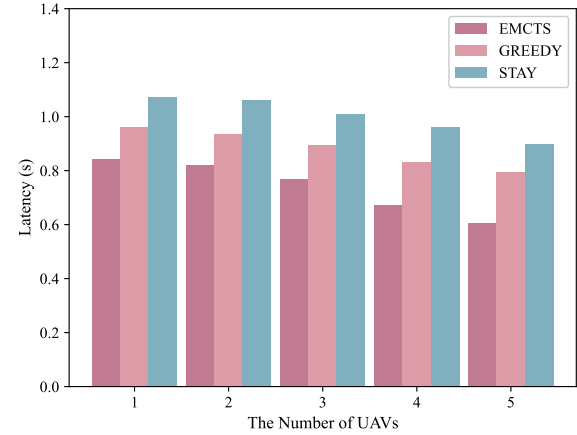


Fig. 8. The average latency for UAV movement strategies with the number of UAVs.

by the dynamic behavior of vehicles in the network. However, our model still exhibits better stability, which indicates the accuracy of the moving area prediction model greatly affects the performance of the caching decision.

C. Evaluation of Cooperative Caching Schemes

Fig. 8 shows the average latency for UAV movement strategies with the number of UAVs. The strategies include EMCTS, GREEDY, and STAY. The GREEDY strategy is that the UAV will always move to the nearby area with the most vehicles. The STAY strategy is that the UAV chooses a served area without changing its position. It can be observed that the average latency decreases as the number of UAVs increases, which means that the UAV can effectively provide content services. This is because the increased number of UAVs expands the system's caching capacity and reduces the communication distance between vehicles and UAVs. Consequently, the UAV can provide service for more vehicles and effectively shorten the content delivery time. Moreover, each UAV only serves vehicles in a small area, so it has a higher channel capacity to provide stable service. On the other hand, EMCTS exhibits a faster decline in average latency compared to GREEDY, especially when the number of UAVs increases. The reason is that the energy-aware scheme can spend more of the overall energy consumption on communication and

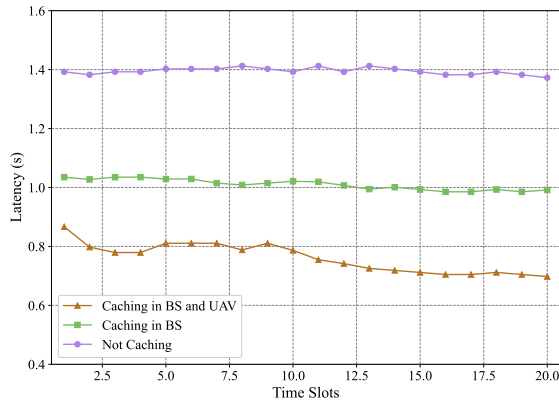


Fig. 9. Content access latency of three scenarios.

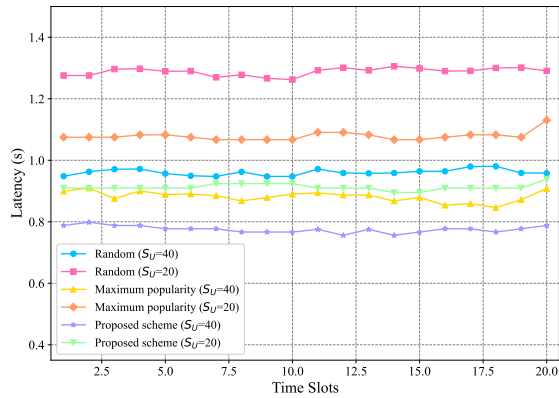


Fig. 10. Content access latency of different caching schemes.

considers additional factors, such as the energy consumption of UAVs, the propulsion cost of UAVs, and the mobility of vehicles.

Fig. 9 shows the average latency of three scenarios: no caching for both UAVs and the MBS, caching only at the MBS, and cooperative caching between UAVs and the MBS. Compared with the case without a cache deployment scheme, the case of caching only at MBS reduces the access latency by 16%, indicating that deploying cache at the edge nodes of IoCV can effectively reduce latency. Moreover, the latency is further reduced by 27% in the cooperative caching scheme. This indicates that employing UAVs for cooperative communication and caching tasks can further enhance latency performance in the case of high vehicle density. Therefore, comparing latency performance under different caching scenarios indicates the meaningfulness of the proposed cooperative caching scheme between UAVs and the MBS, which can effectively improve network performance.

Fig. 10 shows the impact of caching schemes with different cache spaces on latency performance, comparing the random and maximum popularity caching schemes. The random caching scheme is that the MBS and UAVs randomly cache contents, and the maximum popularity caching scheme caches the most popular contents in the network until the cache capacity is reached. The figure shows that the proposed caching scheme outperforms the other schemes, especially the random scheme. The proposed scheme improves by 11.7% compared

to the maximum popularity caching scheme. This indicates that the proposed scheme can more accurately deploy cached contents to improve cache hit rates and reduce latency. This is because predicted content request distribution estimation in the area is obtained by predicting vehicle mobility and the content request preferences of vehicles, and then the local personalized content request distribution is also estimated. In addition, as the cache space increases from 20 MB to 40 MB, the latency of these schemes also decreases due to the number of cached contents increases. The proposed scheme has the smallest latency for $S_U = 40$ MB, which means that increasing the cache capacity can narrow the gap in the latency performance.

V. CONCLUSION

This paper studied UAV-assisted edge caching and trajectory planning in the Internet of Vehicles. In high-speed vehicular scenarios, we formulated a joint optimization problem for the UAV trajectory and edge caching of the MBS and UAVs to minimize network latency. During the process, we comprehensively consider energy consumption, cache capacity, content requests, dynamic network topology, and communication quality. In this work, by analyzing the real data set, it can be observed that it is difficult to obtain real-time vehicle trajectory data, which may be caused by signal transmission delays and dynamic changes in the environment. Moreover, considering that predicting vehicle density distribution does not require real-time trajectory, we proposed to predict the vehicle's future driving area based on the area where it is located during the time interval. Furthermore, the vehicles in the service area will dynamically change as the vehicles move. The node representation learning method based on node features may not be suitable for vehicle moving area prediction. Therefore, we proposed a TBGN model to learn the dynamic network representation. In addition, the distribution of vehicle content requests also changes dynamically with vehicle movement. In order to better describe the dynamic environment, we designed a content request model to describe the content request probability of vehicles. This study provided essential methods and ideas for improving the network performance of edge caching in IoCV.

In future work, we will further design a low-complexity online scheduling algorithm in trajectory planning. Moreover, in the future network, computing nodes and storage nodes will exist everywhere. These nodes have limited resources. AI technology can achieve dynamic decision-making and intelligent resource management. When intelligent model inference is executed at edge nodes, the designed algorithm should have the characteristics of high efficiency, lightweight, and low power while ensuring performance.

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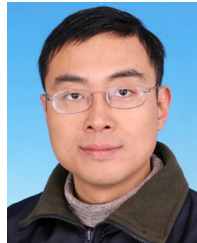


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